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CHEMACT: A Computer Code to Estimate Rate Constants for Chemically-Activated Reactions

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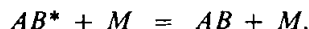
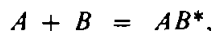
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Abstract—CHEMACT is a computer code that uses the QRRK treatment of chemical activation reactions to estimate apparent rate constants for the various channels that can result in addition, recombination and insertion reactions. The working equations are developed, the necessary input file is described, and sources of required input data are discussed. Three applications of the code to representative examples of reactions relevant to combustion are presented. These include $C_2H_5 + O_2$, $H + C_6H_5Cl$, and $CH_3 + OH$. These reactions illustrate some of the expected behavior with different types of chemical activation and demonstrate the necessity of explicitly accounting for the effects of chemical activation in high temperature gas-phase reactions. Methods by which these results can be included in detailed kinetic mechanisms are discussed.

INTRODUCTION

Kineticists are becoming increasingly aware of the necessity to properly treat the pressure-dependence of simple dissociation and recombination reactions in combustion systems. While the rate constants for many of these reactions involving polyatomic molecules are at the high-pressure limit, *i.e.*, are pressure-independent, at one atmosphere and low temperatures, this limit shifts to much higher pressures as the temperature is increased toward that normally encountered in combustion. These effects are particularly important for recombination or addition reactions where there is a low energy exit channel, *i.e.*, a reaction pathway which has a barrier lower than required for the adduct to reform the initial reactants. Explicit account must be taken of this fall-off behavior when constructing kinetic models for high-temperature systems, as discussed in the review of Warnatz (1983). The shift in the fall-off curves with temperature is simply a manifestation of the fact that these reactions are not simple one-step processes. Consider the recombination reaction:



The shift is due to the markedly different temperature dependence of dissociation and stabilization. The dissociation rate for the energized complex increases rapidly with temperature so that the number of deactivating collisions must be substantially increased to maintain the high-pressure limiting behavior. One would expect to see similar behavior for any reaction which proceeds via a chemically-activated energized complex, *i.e.*, any reaction that involves formation of a new chemical bond. Indeed, a variety of reaction types, including additions, recombinations and insertions, have been analyzed in this way using RRKM theory (Berman *et al.*, 1983; Larson *et al.*, 1984; Golden *et al.*, 1985; Wagner *et al.*, 1990).

Another property of reactions proceeding via energized complexes is the possibility of additional dissociation channels to new products. A particularly interesting example is the $\text{CH}_3 + \text{CH}_3$ reaction. It was shown (Larson *et al.*, 1984) that the products can shift from C_2H_6 (chain termination) to $\text{C}_2\text{H}_5 + \text{H}$ (chain propagation) as the temperature is increased and the complex has enough energy to access the endothermic channel leading to $\text{C}_2\text{H}_5 + \text{H}$. As a result, radical addition, recombination and insertion reactions can manifest very complex temperature and pressure behavior as the stabilization channel competes with these multiple decomposition channels. Substantial errors can result, therefore, when experimental measurements on these rate constants are extrapolated to other regions of temperature and pressure.

A simpler approach to treat these systems, based on QRRK theory and a chemical activation formalism, was devised which permits a more straightforward evaluation of the temperature and pressure behavior of such systems (Dean, 1985). As with the more sophisticated RRKM treatments, it showed good agreement with experimental data and that endothermic channels could become accessible. This QRRK approach has been used to analyze a variety of reactions where there were ample literature data to properly evaluate the predictions of the method (Westmoreland *et al.*, 1986). It was shown to be quite successful in predicting rate constants; in addition, it presented a simple framework by which the effects of temperature and pressure could be readily understood.

The QRRK method is a potent supplement to a RRKM analysis despite being less rigorous. The information required for the QRRK approach is readily available or straightforward to estimate. In particular, the high-pressure “*A*-factor” for dissociation of the energized adduct is based on measured values or assigned by comparison to known rate constants. RRKM theory would predict this *A*-factor from the vibrational frequency distribution and moments of inertia of the transition state. For many of the systems of interest, the structure of the transition state is sufficiently vague that adjustments of these parameters are required to obtain “reasonable” values and thus the predictive power is minimal. Thus, although an RRKM analysis is more accurate in principle, the uncertainties in the input data often restricts its predictive powers, particularly when applied to systems where the available data are sparse.

A shortcoming of the QRRK approach is use of a single geometric-mean frequency to represent the vibrational frequency distribution. One might expect problems with such an approximation, particularly at lower temperatures. Weston (1986) has made extensive comparisons of various formulations of the Kassel approach to RRKM results for a series of thermal unimolecular dissociations. He found that use of a single geometric-mean frequency tends to overestimate $k(E)$ for energies within 10 kcal/mole of the critical energy for dissociation; the overestimate is worse when the arithmetic-mean frequency is used. Comparison of predictions of fall-off behavior showed that the deviations from RRKM predictions are reaction-specific, but it was not possible to correlate the deviations with specific properties of the reactant. In any event, the average deviation from the RRKM predictions was approximately a factor of 2.4. Our recent analysis (Bozzelli *et al.*, 1990) revealed quite good agreement to data over a range of temperatures and pressures, including low temperature regimes.

Another advantage of using the QRRK theory is the ease of formulation for cases where the energized adduct has multiple exit channels (Dean, 1985). Thus it appears that the QRRK approach represents a suitable compromise for many cases of interest. It allows one to estimate the temperature and pressure dependence of many reactions for inclusion in kinetic models. Prudence dictates that reactions which are critical in such a model be examined more carefully by use of RRKM analysis and/or master equation methods. However, the QRRK approach allows one to take the first steps

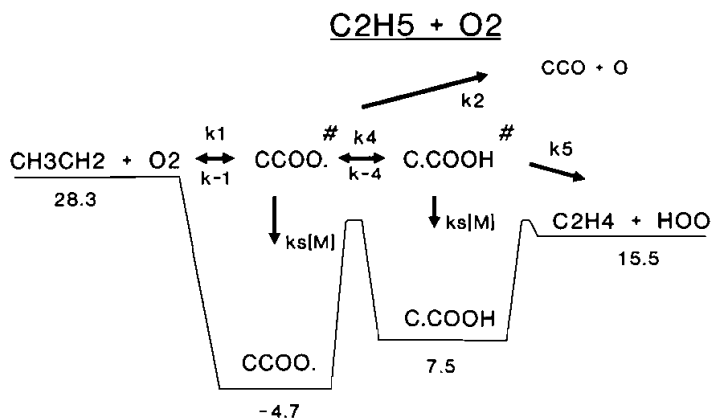


FIGURE 1 Potential energy diagram for the addition of C₂H₅ to O₂. The numbers are the enthalpies in kcal/mole associated with the various species.

toward a more realistic description of the temperature and pressure dependence of addition, insertion and recombination reactions.

In this paper we will describe a computer program CHEMACT, based on the analysis of Dean (1985), that can be used to estimate the pressure and temperature dependence of the various channels of reactions that proceed through formation of a chemically activated complex. Several examples will be given that both illustrate use of CHEMACT and describe systems of interest in combustion chemistry.

DEVELOPMENT OF THE WORKING EQUATIONS

Consider the reaction sequence for addition of C₂H₅ to O₂ shown in Figure 1. The energized complex CCOO* can either dissociate, isomerize or be collisionally stabilized (the stabilization rate constant is the product of the collisional stabilization rate constant and the deactivation efficiency β). The dissociation rate constants k_{-1} , k_2 and the isomerization rate constant k_4 are functions of energy. Similarly, the excited isomer C'COOH* can dissociate (k_5), isomerize (k_{-4}) or be collisionally stabilized. Applying the steady-state assumption to CCOO* and C'COOH*, one can define the following "apparent" rate constants:

$$k_{\text{stab}_A} = \frac{d[\text{CCOO}^*]/dt}{[\text{C}_2\text{H}_5][\text{O}_2]} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{\beta k_s[M]N}{D} f(E), \quad (1)$$

$$k_{\text{rxn}_A} = \frac{d[\text{CCO}^*]/dt}{[\text{C}_2\text{H}_5][\text{O}_2]} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{k_2 N}{D} f(E), \quad (2)$$

$$k_{\text{stab}_B} = \frac{d[\text{C}'\text{COOH}]/dt}{[\text{C}_2\text{H}_5][\text{O}_2]} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{\beta k_s[M]k_4(E)}{D} f(E), \quad (3)$$

$$k_{\text{rxn}_B} = \frac{d[\text{C}_2\text{H}_4]/dt}{[\text{C}_2\text{H}_5][\text{O}_2]} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{k_4(E)k_5(E)}{D} f(E), \quad (4)$$

where

$$N = [k_{-4}(E) + \beta k_s[M] + k_5(E)]$$

and

$$D = N[k_{-1}(E) + \beta k_s[M] + k_2(E) + k_4(E)] - k_4(E) \cdot k_{-4}(E).$$

In these equations, $f(E)$ is the fraction of $\text{CCOO}^{\cdot*}$ or $\text{C}^{\cdot}\text{COOH}^{\cdot*}$ at a specific energy E , the allowed energy levels are in increments of $h\nu$ (ν = geometric mean frequency), and only energies above E_{crit} , which is the threshold energy for the specific reaction, are considered. $f(E)$ is defined as the chemical activation distribution (Robinson *et al.*, 1972), and it can be expressed as:

$$f(E) = \frac{k_{-1}(E) \cdot K(E)}{\sum_{E_{\text{crit}}}^{\infty} k_{-1}(E) \cdot K(E)}, \quad (5)$$

with

$$K(E) = \alpha^n (1 - \alpha)^s \frac{\Gamma(n + s)}{\Gamma(n + 1)\Gamma(s)}, \quad (6)$$

and

$$k_i(E) = A_i \frac{\Gamma(n + 1)\Gamma(n - m_i + s)}{\Gamma(n - m_i + 1)\Gamma(n + s)}, \quad (7)$$

where A_i is the high pressure Arrhenius pre-exponential factor for reaction i , m_i is the number of quanta ($E_i/h\nu$) corresponding to the energy threshold for reaction i , $\alpha = \exp(-h\nu/kT)$, s = number of vibrational degrees of freedom, and $n = E/h\nu$, *i.e.*, the number of quanta of energy. Our current modification to the formulation presented by Dean (1985) affects only the evaluation for the ratio of excitation to deexcitation rate constants [$K(E)$] and the energy dependent rate constants [$k_i(E)$]. These modifications amount to replacing factorial evaluations with Gamma function evaluations. Factorial evaluations require that m_i , the number of quanta which correspond to the energy threshold for reaction to channel i ($m_i = E_i/h\nu$), be rounded to the nearest integer. Since $E_i/h\nu$ typically does not have an integer value, the Gamma function treatment avoids this approximation. The critical number of quanta m_i for $k_{-1}(E)$, $k_2(E)$, and $k_4(E)$ are calculated from the bottom of the first potential well [$\text{CCOO}^{\cdot}$], while that for $k_{-4}(E)$, and $k_5(E)$ are based on the bottom of the potential well of the isomerized complex [$\text{C}^{\cdot}\text{COOH}$]. Summations with respect to energy are carried out between the energy which corresponds to the threshold energy for dissociation of the complex to reactants, E_{crit} , and infinity. Summations are therefore performed from $n = n_{\text{crit}} = E_{\text{crit}}/h\nu$ to ∞ , with $n = n_{\text{crit}} + I$, where I is an integer corresponding to the energy level above n_{crit} . Here again n_{crit} is typically not an integer. Gamma function evaluations were performed using the Stirling asymptotic series which approximates the Gamma function for values of $n > 1$ (Beyer, 1983). This Gamma function treatment is expected to give more accurate estimates at lower temperatures, where small changes in barrier may have significant effects. (The above

equations are only shown for the channels in this specific example of $\text{C}_2\text{H}_5 + \text{O}_2$, but the CHEMACT code allows two dissociation channels from the first complex (k_2 and k_3) and two from the second (k_5 and k_6).

The values of the collisional stabilization rate constant (k_s) are computed using Lennard-Jones collision rates (Forst, 1973):

$$k_s = 2.708k_{hs}(\varepsilon/k_B T)^{1/3}, \quad (8)$$

where k_{hs} is the usual hard-sphere collision rate constant. Collision efficiencies (β) were evaluated using the expression (Troe, 1979):

$$\frac{\beta}{1 - \beta^{1/2}} \cong \frac{-\langle \Delta E \rangle}{F_E k_B T}. \quad (9)$$

For this work, F_E was assumed to be 1.15 (cf. Table II of (Troe, 1979)). Other values of β can be used in the calculations by specifying them in the input file on Line 52 (cf. below).

Treatment of Collisional Deactivation

A possible problem with use of βk_s in the context of chemical activation is that values of β were deduced for thermal activation systems where only one dissociation channel was considered as opposed to the multiple chemical activation channels considered here. Troe's approach allows one to use a modified strong collision assumption in that every collision is not deactivating but whenever deactivation does occur it is complete, *i.e.*, all the excess energy is lost. The decrease in β with increasing temperature then allows for an energy distribution represented by somewhat high excitation levels with correspondingly lower probability of deactivation to a level below the critical energy. The introduction of a second dissociation channel complicates this issue since it is now possible for the energized complex to lose sufficient energy so that one channel is not accessible while the second reaction (at a slower rate) could continue through the lower energy channel. Thus use of βk_s (which "kills" both channels) will introduce a bias in the result when the energy gap between channels is larger than the energy transferred per collision (Larson *et al.*, 1969). For hydrocarbon systems, $-\langle \Delta E \rangle \sim 2$ kcal/mole (Gardiner *et al.*, 1984) and differences in exit channel barriers much greater than this could cause problems.

However, comparisons to master equation solutions indicate the use of β can often be acceptable (Dean, 1985). For many cases, the various reaction channels are comparable in energy and the complications do not arise. Moreover, in cases involving widely different barriers, the lower energy barrier is often sufficiently low, relative to the energy of the chemically activated adduct, that the adduct will dissociate on a very fast timescale, since the unimolecular dissociation rate is so dependent on the internal energy. This fast dissociation rate can be much faster than the collisional deactivation process. In this case, the calculation is insensitive to the details of the energy transfer process. Nevertheless, the user of CHEMACT should be aware of potential pitfalls resulting from use of the simplified energy transfer approach. Troe (1983) has presented an analysis for chemical activation for a single channel case, while Oref and Tardy (1990) discuss approaches for solution of the master equation.

DESCRIPTION OF THE INPUT DATA FOR CHEMACT

Table I is a reproduction of the input file required to run the CHEMACT program for the reaction of $\text{C}_2\text{H}_5 + \text{O}_2$. The following text gives a line-by-line description of

TABLE I
Recombination and decomposition (with isomerization) using QRRK

<pre> P2 P3 P5 / \ / \ / K1 K2 / K3 K4 K5 ===== A* ===== B* ----- P6 K-1 K-4 A B </pre>						1
						2
						3
						4
						5
						6
						7
						8
						9
						10
						11
						12
						13
R (REACTANTS)						14
CC. + O2						15
A* (RECOMBINATION) P2 (1ST A* DECOMP) P3 (2ND A* DECOMP)						16
CCOO. CCO + O						17
B* (ISOMER STABIL) P5 (1ST B* DECOMP) P6 (2ND B* DECOMP)						18
C.COOH C=C + HO2						19
FREQ (FREQUENCY, CM ⁻¹) NS (Number Oscillators)						20
1081. 21						21
HIGH PRESSURE RATE CONSTANT: A (M-CC-S) N E (KCAL)						22
3.0E+12, 0.0, 0.0						23
COLLISION COMPLEX MASS(A.U.) DIAMETER (A) WELL DEPTH (K)						24
61., 4.36., 450.0						25
THIRD BODY						26
N2 MASS (A.U.) DIAMETER (A) WELL DEPTH (K) ENERGY TRANS (CAL)						27
28., 3.62, 97.5, 830.1						28
A(CC/Mole-Sec) E (Kcal) k-1						29
1.89E15, 32.04, k-1						30
1.4E15, 59.6, k2						31
0.0, 0.0, k3						32
2.6E12, 27., k4						33
1.45E11, 15., k-4						34
5.5E12, 15.6, k5						35
0.0, 0.0 k6						36
DELTA H FOR A* ==> B* (KCAL/MOL) BETA (Collision Deactivation coeff.)						37
12., 0.0 (calculated if = 0.0)						38
NUMBER OF TEMPERATURES T1 T2 T3 T4 T5 T6 (K)						39
5, 300., 500., 800., 1000., 1500.,						40
NUMBER OF PRESSURES P1 P2 P3 P4 P5 P6 (TORR)						41
1, 760., , , , ,						42
Tdist (FOR DISTRIBUTION Ki(E) vs. E AT THIS TEMP (K))						43
500.						44
Pdist (for distribution Ki(E) vs. E at this Press (Torr))						45
760.						46
OUTPUT OPTION (-1,0,1,2,3,4,5, or 6)						47
6, OPTION - All options include 2 & 3 parameter ARRHENIUS fit of QRRK results						48
OUTPUT OPTIONS :						49
-1 - CHEM ACT DISTRIBUTION FN & (ki/kinf) AT T= Tout and P=Pout only						50
0 - CHEM ACT DISTRIBUTION FN & (ki/kinf) ALL T & P						51
1 - CHEM ACT DISTRIBUTION FN & (ki/kinf) AT P=Pout and ALL T						52
2 - CHANNEL SUMMARY (ki/kinf) AT ALL TEMPS & PRESSURES						53
3 - CHANNEL SUMMARY (ki/kinf) at T=Tout & P=Pout ONLY						54
4 - (ki/kinf) at Tout and all P (ASCII output for FALL-OFF plot)						55
5 - (ki/kinf) at Pout and all T (ASCII output for ARRHENIUS plot)						56
6 - output no CHEM ACT DISTRIBUTION FN or CHANNEL SUMMARY						57
(only 2 & 3 parameter ARRHENIUS fits)						58

this input:

Lines 1–13: Descriptions of the reaction channel for which the code will calculate apparent rate constants. (No data is read in on these lines.) R represents reactants; P_i , $i = 2, 3, 5$ or 6 represents products; A^* represents the initially formed energized complex; A represents the stabilized complex; B^* represents an energized isomer of complex A^* and B represents the stabilized isomer. k_1 represents the high pressure limiting rate constant for reactants to form the stabilized complex A ; k_{-1} represents the reverse reaction from the energized complex A^* back to reactants. It should be noted, however, that the input to CHEMACT is the rate constant for dissociation of the stabilized adduct A (as opposed to the energized adduct A^*). This approach allows k_1 and k_{-1} to be related via the equilibrium constant for formation of the stabilized complex, which is easily computed (cf. Appendix I). We strongly recommend that either k_1 or k_{-1} be calculated in this way so that thermodynamic consistency is maintained. k_2 and k_3 represent reaction from A^* to product channels 2 and 3. Again it is noted that the input parameters to the CHEMACT are based on reaction from A . k_4 is the rate constant for isomerization of A^* to B^* . As above, this is based upon reaction of A , not A^* . k_{-4} is the reverse isomerization rate constant; again we recommend that either k_4 or k_{-4} be calculated using the equilibrium constant. k_5 and k_6 represent the rate constants for reaction of B^* to product channels 5 and 6, with the rate parameters being those for reaction from the stabilized complex.

(Data in lines 15, 18, and 21 is alphanumeric and is only used for printout, not in any calculations.)

Line 15: Columns 1–16 accepts identification of the reactants

Line 18: Columns 1–16, 21–36 and 41–56 accepts identification of the initial adduct A and decomposition products for channels 2 and 3.

Line 21: Columns 1–16, 21–36 and 41–56 accepts identification of isomer B and decomposition products 5 and 6 from this isomer.

(Data are input on lines 24, 27, 30, 34, 37, 39, 41, 43, 45, 47, 49, 52, 56, 58, 64, and 67. These data are free format and need only be separated by commas.)

Line 24: Frequency in cm^{-1} , number of vibrational modes. Caution: CHEMACT cannot be used for the case of a diatomic adduct where there is only one oscillator.

Line 27: k_1 in $\text{cm}^3/\text{mole-sec}$ in the form of $A \cdot T^n \cdot \exp(-E_a/RT)$, input A , n , and E_a in kcal/mole

Line 30: Molecular parameters for complex A : mass (amu), collision diameter (A) and well depth (ϵ/k_B) in Kelvin

Line 34: Molecular parameters for the bath gas: mass (amu), collision diameter (A) and well depth (ϵ/k_B) in Kelvin, average energy transferred per collision in cal/mole .

Line 37: 39, 41, 43, 45, 47, and 49: Rate constants k_{-1} , k_2 , k_3 , k_4 , k_{-4} , k_5 and k_6 in Arrhenius form $A(\text{sec}^{-1})$, $E_a(\text{Kcal/Mole})$

Line 52: ΔH_{rxn} for isomerization $A \rightarrow B$ (kcal/mole) and beta, which is an input control for calculation of β (the fraction of collisions which stabilize of the energized adduct A^* and B^*). If $\beta = 0$, CHEMACT will calculate β according to Troe's formalism (cf. text). Otherwise, β is input directly here. (β must be less than 1.0.)

Line 55: Number of temperatures and the specific temperatures (K) for which calculations are to be performed. The code performs calculations for up to 6 temperatures; at least two are required for Arrhenius fits and 3 for modified Arrhenius fits.

Line 58: Number of pressures and the specific pressures (Torr) for which calculations are to be performed. The code performs calculations for up to 6 pressures.

Lines 61 and 64 allow specification of a temperature and pressure for output of additional information in the form of specific rate constants $[k_i(E)]$ and fraction of energized complexes $[f_i(E)]$ at energy level E_i .

Line 67: Output option—choose values for $-1, 0, 1, 2, 3, 4, 5$, or 6 . This option controls the specific data and format of the output. The available options are listed in Table I.

(Other lines in the input file are not used to input data. They are either descriptive comments or blank spacer lines. The latter may be used to add descriptive comments, such as references to rate constants used, etc.)

GENERAL SOURCES OF INPUT DATA FOR CHEMACT

I *The thermodynamic parameters*, enthalpy (ΔH_f^0), entropy (ΔS_f^0), and heat capacities (C_p) as a function of temperature for the reactants, adducts and products of the reaction system are important for accurate results. (The superscript in the above denotes a standard state of 298 K.) These parameters are critical for accurate determination of the following:

—The well depth, which is the excess energy of the initially formed complex, due to formation of the new chemical bond

—Energies of activation for unimolecular bond cleavage reactions of the energized complex

—Determination of rate constants for various reaction channels through principles of microscopic reversibility. Appendix A presents a summary of the necessary calculations and they are also readily performed as a function of temperature in the THERM computer code (Ritter *et al.*, 1991).

These thermodynamic parameters are available from a number of extensive tables in the literature. These include data published by the Thermodynamics Research Center (Rogers, 1982), the text by Stull, Westrum and Sinke (1987), the Chemkin Thermodynamic Data Base (Kee *et al.*, 1987), and Lias *et al.* (1988). The thermodynamic data can also be calculated by Group Additivity methods of Benson (1976), or THERM (Ritter *et al.*, 1991), a computer code which utilizes the Benson Group Additivity scheme. An added source for stable species heats of formation at 298 K is the evaluated data compiled by Pedley, Naylor and Kirby (1987). Bond energies for radicals can be obtained from evaluations of current literature as a number of studies are ongoing in this field (Tsang, 1978; McMillen *et al.*, 1982; Tsang, 1985; Tsang *et al.*, 1986; Gutman *et al.*, 1988; Gutman *et al.*, 1988; Holmes *et al.*, 1988; Holmes *et al.*, 1988; Tschjuikow-Roux *et al.*, 1989; Ervin *et al.*, 1990; Tschjuikow-Roux *et al.*, 1990). Bond energies can also be estimated using calculational methods (Leroy *et al.*, 1989; Stein *et al.*, 1991), but we recommend comparison of these results with experimental data wherever possible. It is apparent that heat of formation calculations for radicals may have significantly more error than calculations for stable closed shell molecules.

II. *Molecular parameters* describing the size, collisional energy transfer and energy levels of the adduct formed by the initial reaction are also needed. These include:

—The mass and the number of vibrational modes of the adduct ($3N - 6$, for nonlinear molecules, where N is the number of atoms in the complex, and $3N - 5$ for linear species).

—The Lennard-Jones parameters, σ and ϵ . σ is the collision diameter in angstroms, and ϵ/k_B is the well depth in Kelvin. These parameters are available from a number of sources: Values for species frequently encountered in combustion applications are

listed in Kee *et al.* (1983). A fairly extensive source is the current edition of Reid, Prausnitz and Poling (1987). This text (RPP) also presents a method to estimate σ and ϵ/k_B from the critical volume (V_c) and temperature (T_c) respectively (Chung, 1984):

$$\sigma = 0.809 \cdot V_c^{1/3},$$

$$\epsilon/k_B = T_c/1.259,$$

with substantial tabulations of these critical parameters tabulated in their Appendix as well. RPP includes an updated method to estimate these critical properties using increments derived by Joback and Lydersen Groups. Herschbach and Ben Amotz (1990) also present tabulations for a number of hydrocarbon species. A second method for estimating the critical parameters by the Benson Group additivity approach is described by Jawalka and Daubert (1986; 1989). This method also requires the boiling point of the compound, however. The Lennard-Jones parameters for radical complexes are estimated by averaging two structurally similar molecules, with combining rules presented by Prausnitz (1986). For example, cyclohexadienyl radical C_6H_7 ($CHD^\cdot$) can be estimated by:

$$(\epsilon/k_B)_{CHD^\cdot} = [(\epsilon/k_B)_{benzene} \times (\epsilon/k_B)_{CHD}]^{1/2},$$

and

$$\sigma_{CHD^\cdot} = [\sigma_{benzene} + \sigma_{CHD}]/2.$$

The geometric mean vibrational frequency of the adduct is the remaining parameter needed and it is relatively important. This can be calculated from tables of vibrational frequencies compiled by Jacox (1984; 1990) or Shimanouchi (1972). The geometric mean frequency can also be determined using a computer code CPFIT (Ritter, 1989) which accepts input in the form of heat capacities versus temperature to 1000 K, in addition to the number of vibrational modes and the number of internal rotors in the molecule. This code fits the heat capacity data in the above range to a five parameter harmonic oscillator model and extends the temperature range to 5000 K. One additional method of estimating the geometric mean frequency of a radical is to take the arithmetic mean of the corresponding stable molecules with one more H atom ("parent" molecule) and with one less H atom than the radical of interest. Examples would be ethane and ethylene for the ethyl radical, and ethanol and acetaldehyde for the ethoxy radical.

III. *The bath-gas molecules* collision diameter (σ) well depth, (ϵ/k_B) and average energy transferred per collision are also needed. Values of the average energy transferred ($-\langle \Delta E \rangle$) were taken from Gardiner and Troe (1984). Table II lists these parameters for common bath gas species: argon, carbon dioxide, helium, hydrogen, methane, nitrogen, oxygen, and water vapor. A much more comprehensive compilation of energy transfer parameters has recently been published (Oref *et al.*, 1990).

IV. *High pressure limit rate constants* for adduct formation and for various isomerization and dissociation product channels of the adduct are also needed. (This includes dissociation back to reactants.) The rate constants are taken from literature data, if available, along with known thermodynamic properties for each species and transition state in the reaction scheme. The rate constants can be estimated, if literature data is not available, from generic analysis of the reaction (comparison of

TABLE II
Molecular properties for common bath gas species¹

Molecule	Mass (amu)	σ (Å)	ϵ/k_B (K)	ΔE_{avg} (cal/mole)
He	4	2.581	10.2	431
He	4	3.16 ²	3.77 ²	407 ³
Ar	40	3.33	136.5	630
H ₂	2	2.827	59.7	621
N ₂	28	3.62	97.5	830
O ₂	32	3.458	107.4	971
CH ₄	16	3.75	141.4	2100
H ₂ O	18	2.605 ⁴	572 ⁴	1425
CO	28	3.654 ⁴	98.1 ⁴	
CO ₂	44	3.763 ⁴	244 ⁴	

¹Gardiner *et al.* (1984), unless otherwise specified.

²Reid *et al.* (1987).

³Pilling and Lightfoot as listed in Gardiner *et al.* (1984).

⁴Kee *et al.* (1983).

the reaction to a similar reaction system, where the rate parameters are known) or from Transition State Theory (Benson, 1976). We note that estimation is just that, an estimate, and experimental data are always preferable to make the kinetic calculations more accurate. Some typical literature tabulations include the NIST Data Base (Mallard, 1990), Tsang (1986; 1987; 1988; 1990), Cohen (1983; 1989; 1990), Westley *et al.* (1987), Atkinson (1989) (for OH radical abstraction reactions), Baulch *et al.* (1973), Cvetanovic (1987) and Herron (1988) (reactions of oxygen atoms), Troe (Borrel *et al.*, 1985) (high pressure limit rate constants), Allara and Shaw (1980), Kerr and Moss (1981) and Kerr and Drew (1987). Gardiner's text includes tabulations for hydrocarbon oxidation (Warnatz, 1984) and nitrogen reactions (Hanson *et al.*, 1984). (Rate constants for stabilization of the energized adduct are computed as described earlier.)

DESCRIPTION OF TYPICAL OUTPUT

Table III shows portions of the output file generated using the input shown in Table I. The initial portion of the output, which mirrors the input file, is omitted. The first block of output shown lists the normal Arrhenius fit [$k = A \exp(-E/RT)$] to the calculated apparent rate constants for the first channel (stabilization). The second block lists the corresponding modified Arrhenius fit [$k = AT^n \exp(-E/RT)$] for this same channel. The program has been designed to shrink the temperature range of the modified Arrhenius fit, if necessary, to achieve a average fit of better than 10%. Note this occurred in both reactions shown in Table III. In general, one finds a much better fit using the modified Arrhenius form, even when the temperature range is identical for both fits. This sequence is then repeated for the additional reaction channels. For brevity, output is only shown for one other channel; in this case, it is the dissociation of the isomerized adduct to give ethylene and HO₂. (If the input specified multiple pressures, the above material would be repeated for each pressure in turn.)

TYPICAL APPLICATIONS

Alkyl Radicals + Oxygen

There has been considerable disagreement in the literature regarding these reactions.

Most of the controversy revolves around the possibility of an intramolecular hydrogen shift to convert the originally formed peroxy radical (Figure 1) to an alkyl hydroperoxy radical followed by dissociation. If the shift does occur, the ultimate result of the addition/shift/elimination can be equivalent to a simple hydrogen transfer, but, of course, the effects of temperature and pressure could be dramatically different depending upon the actual path the reaction takes. QRRK analyses have been done for both the methyl (Dean *et al.*, 1987) and ethyl (Bozzelli *et al.*, 1990) additions. The ethyl case will be considered here as an example of a typical application.

The input parameters are referenced to the (stabilized) ground state of the complex because this is the formalism used in QRRK Theory (Kassel, 1928a; Kassel, 1928b). The ethyl radical combines with O_2 to form the chemically activated $CCOO^{\cdot\#}$ adduct. The major reaction channels of $C^{\cdot}COOH^{\#}$ can then:

- (1) Dissociation back to reactants (k_{-1}).
- (2) Stabilization to $CCOO^{\cdot}$ (βk_s).
- (3) Dissociation to $CCO^{\cdot} + O$ (k_2).
- (4) Isomerization via a 5-member cyclic intermediate to $C^{\cdot}COOH^{\#}$ (k_4).
- (5) Beta scission to $C_2H_4 + HO_2$ (k_5).
- (6) Be stabilized to $C^{\cdot}COOH$ (βk_1).
- (7) Or isomerize back to $CCOO^{\cdot}$ (k_{-4}).

The thermodynamic parameters for most species important to the potential energy level diagram and those used in the mechanism to follow the reaction progress with time (cf. (Bozzelli *et al.*, 1990) for details on use of the mechanism) are listed in Appendix B. Standard thermodynamic parameters were used for all other species. C_2H_5 thermodynamic properties are taken from the literature (Brouard, 1986; Tschuikow-Roux, 1989; Tschuikow-Roux, 1990). $C_2H_5OO^{\cdot}$ is based on the parent ethyl hydroperoxide C_2H_5OOH and a $ROO-H$ bond energy of 88 kcal/mole (Baldwin, 1983) with entropy and heat capacity parameters also reported by Baldwin (1983) and by Wagner *et al.* (1990). The $C^{\cdot}COOH$ radical thermodynamics considers the radical site to be a primary radical with a C-H bond similar to the ethyl radical and used the ethyl hydroperoxide parent as reference. Ethylene and HO_2 are taken from the literature (Rogers, 1982; Chase *et al.*, 1985).

The parameters describing the adduct $C_2H_5OO^{\cdot}$ were determined as described earlier; LJ parameters were taken from Herschbach and Ben Amotz (1990) for the molecule propanol, $CCCOH$. The geometric mean vibrational frequency of the adduct was taken directly from calculated frequencies by Wagner *et al.* (1990). This value was different by only 35 cm^{-1} from that calculated by our computer code CPFIT (Ritter, 1989). The cross sectional diameter, well depth and average energy transferred per collision are shown in Table II.

The remaining input to the CHEMACT is the kinetic high pressure limit rate constant for reaction to the adduct and for the various reaction, isomerization and stabilization channels of the adduct. The stabilization reaction rate constants are determined from Troe's formalism and the molecular parameters describing the adduct as listed above. k_1 was taken from the literature (Monk *et al.*, 1986). k_{-1} was then calculated as the average rate constant for the reverse reaction over the temperature range 300 to 1200 K using microscopic reversibility and THERM (Ritter *et al.*, 1991). The dissociation of the adduct to ethoxy + O (k_2) was estimated from the reverse reaction. The rate constant for combination of oxygen and ethoxy was taken as 0.55 that for $CH_3 + O$. The value for $CH_3 + O$ was taken from Washida and Bayes (1980a; 1980b) and the value of 0.55 is the steric factor taken from Benson (1983). The barrier in the isomerization of $CCOO^{\cdot}$

TABLE III
Examples of output data from CHEMACT code

2 Parameter Arrhenius for apparent rate constants at $P = 7.60E+02$ Torr $C_2H_5 + O_2 = CCOO$, T 300 K to 1500 K			
$A = 4.92E+10$		$E = -2.93E+00$ Kcal	
T	K_i {qrrk}	K_i {fit}	Percent error
300	$2.96E+12$	$6.76E+12$	-56.1
500	$2.59E+12$	$9.43E+11$	174.5
800	$1.00E+12$	$3.12E+11$	221.7
1000	$3.57E+11$	$2.15E+11$	65.8
1500	$2.05E+10$	$1.32E+11$	-84.4
3 Parameter Arrhenius for apparent rate constants at $P = 7.60E+02$ Torr $C_2H_5 + O_2 = CCOO$, T 300 K to 1500 K Non-linear regression results $9.657E+50 * T^{-12.22} * \exp(-12.81 \text{ Kcal}/RT)$			
Temp (K)	K_i (qrrk)	K_i (fit)	Percent error
500	$2.58E+12$	$2.57E+12$	-.59
800	$1.00E+12$	$1.04E+12$	3.90
1000	$3.57E+11$	$3.42E+11$	-4.23
1500	$2.05E+10$	$2.07E+10$	1.10
2 Parameter Arrhenius for apparent rate constants at $P = 7.60E+02$ Torr $C_2H_5 + O_2 = C=C + HO_2$, T 300 K to 1500 K			
$A = 1.26E+11$		$E = 3.31E+00$ Kcal	
T	K_i {qrrk}	K_i {fit}	Percent error
300	$3.62E+08$	$4.86E+08$	-25.5
500	$6.03E+09$	$4.49E+09$	34.4
800	$2.71E+10$	$1.57E+10$	72.7
1000	$2.95E+10$	$2.38E+10$	23.9
1500	$1.94E+10$	$4.15E+10$	-53.3
3 Parameter Arrhenius for apparent rate constants at $P = 7.60E+02$ Torr $C_2H_5 + O_2 = C=C + HO_2$, T 300 K to 1500 K Non-linear regression results $5.089E + 29 * T^{-5.61} * \exp(-10.92 \text{ Kcal}/RT)$			
Temp (K)	K_i (qrrk)	K_i (fit)	Percent error
500	$6.03E+09$	$6.04E+09$	.59
800	$2.70E+10$	$2.66E+10$	-1.69
1000	$2.94E+10$	$3.00E+10$	1.94
1500	$1.93E+10$	$1.92E+10$	-.49

to $C^{\cdot}COOH$ is based on a combination of ring strain and the activation energy for abstraction of a H from the $-CH_3$ group by an $HOO^{\cdot}$ radical. The contribution from ring strain is estimated from Benson's group additivity ring correction tables (Benson, 1976). In this case we use the ring correction for the five member ring with 4 carbons and 1 oxygen (5.9 kcal/mole). If this heteroatom data were not available, a cyclic C_5 ring strain could be used (6.3 kcal/mole). The activation energy for the abstraction, which is 12 kcal/mole endothermic, is estimated to be approximately 21 kcal/mole, *i.e.*, 9 kcal/mole above the reaction endothermicity. This value is based upon analysis of exothermic abstraction reactions and the use of microscopic reversibility. Thus, the overall $E_{isomerization}$ is then estimated to be $6 + 21 = 27$ kcal/mole.

The A -factor for the isomerization is based on Transition State Theory (Benson, 1976):

$$A = g(e k_B T/h) \exp(\Delta S^\ddagger/R),$$

where $\Delta S^\ddagger$ is calculated from loss of 2 rotors, gain of an optical isomer, loss of H-C-C and C-O-O bends, loss of C-H stretch, gain of C-H-O and H-O-O- bends. The degeneracy factor g is used to account for the number of hydrogen atoms that can be transferred. $\Delta S^\ddagger$ was calculated to be -7.5 cal/K-mole. Note use of g implies that the entropy of activation *not* include the change of symmetry. [An alternate method is to include this symmetry in the entropy and set the degeneracy factor to unity (Laidler, 1969).] The C-OO rotor is treated as C_2H_5 . We have developed a generic set of $\Delta S^\ddagger$ values for loss of rotors in these reactions. While we recommend that one use a more specific TST calculation, the generic values may be somewhat easier to use in performing estimates of the rate constants. The $\Delta S^\ddagger$ values track with number of rotors lost and are summarized in the table below.

Number of Rotors lost	$\Delta S^\ddagger$
1	-4.0
2	-7.5
3	-10.5
4	-13

The activation energy for addition of HO_2 to ethylene, which is the reverse reaction dissociation of C^+COOH , was initially determined from a literature analysis to be 8 kcal/mole, similar to hydrocarbon radical additions to ethylene. The final value of 8.3 kcal/mole was chosen for optimum fit to the data of Kaiser *et al.* (1989; 1990). A_{-5} was taken from an average of values reported in Kerr and Moss (1981) to be 5.5×10^{11} . We note that this is similar in value (slightly higher) to hydrocarbon radicals, *e.g.* CH_3 addition to olefins, and we feel addition of HO_2 to olefins is similar to addition of the much more highly studied hydrocarbon radicals.

CHEMACT uses this input to estimate rate constants for each of the channels at a variety of temperatures and pressures. Typically all of the channels will exhibit some pressure dependence. Generally, stabilization channels exhibit "fall-off" behavior, where the rate constant scales with pressure at low pressures and becomes independent of pressure at higher pressures, with the transition being a strong function of temperature. The dissociation channels typically show inverse fall-off behavior (Westmoreland *et al.*, 1986), where the rate constant varies inversely with pressure at high pressure and is independent of pressure at low pressures, again with the transition region being a function of temperature. A complete discussion of the results for $C_2H_5 + O_2$ is given elsewhere (Bozzelli *et al.*, 1990), but the next section covers some of the highlights.

A plot of the apparent rate constant to ethylene + HO_2 versus temperature for three pressures (N_2 bath gas) is illustrated in Figure 2. The data show that at lower pressures more of the complex reacts to $C_2H_4 + HO_2$. We note that the temperature dependence for the apparent rate constant to the C_2H_4 channel depends on the pressure. At low pressure, the small decrease is due to the higher probability for dissociation back to reactants at higher temperature (much higher A factor for dissociation). At higher pressures, although the rate is lower since there is now more stabilization, we see an increase in rate with temperature. For comparison, these results are compared to the "conventional" picture which assumes a direct hydrogen transfer reaction to form ethylene and HO_2 . Note the much more complex behavior

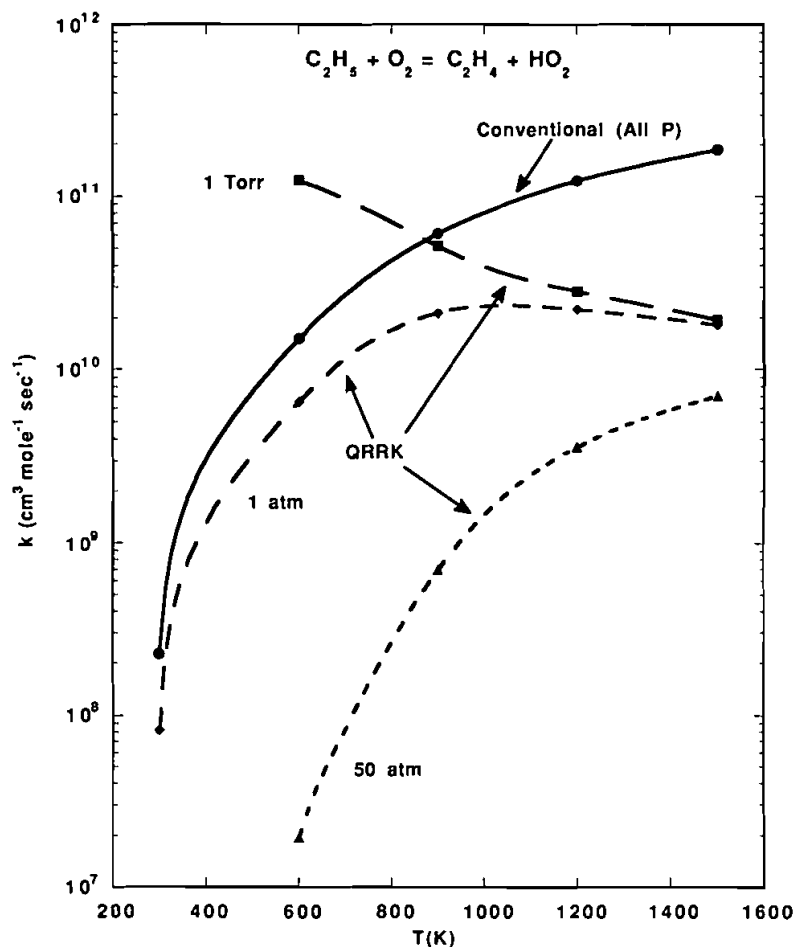


FIGURE 2 Predicted temperature dependence for the reaction $\text{C}_2\text{H}_5 + \text{O}_2 = \text{C}_2\text{H}_4 + \text{HO}_2$ at several pressures. The various dashed lines are the QRRK results at the specified pressures, while the solid line is the pressure-independent value which assumes the reaction is a simple hydrogen-transfer rather than one which proceeds via an energized complex [from W. J. Pitz and C. K. Westbrook, *Comb. Flame* **63**, 113–33 (1986)].

predicted using the energized complex formalism. This comparison illustrates the dangers inherent in extrapolating a rate constant into regions of temperature and/or pressure that are far away from those where the value was measured.

A plot of apparent rate constants versus pressure for two temperatures, 300 and 1200 K, is shown in Figure 3. We again point out that dissociation of the complex to $\text{C}_2\text{H}_5 + \text{O}_2$, the reverse reaction, is the primary reaction channel of the energized adduct at low pressure. At 300 K, stabilization to $\text{CCOO}^\bullet$ approaches the high pressure limit at 1 atm pressure, but at 1200 K stabilization is still well into the falloff regime at 1 atm. Reaction to C_2H_4 at 300 K shows a maxima at low P and steadily decreases with increasing pressure. At 1200 K, however, the ethylene yield is nearly constant, resulting from dissociation to ethyl being the dominant channel and stabilization not being important through the pressure range up to 1 atm.

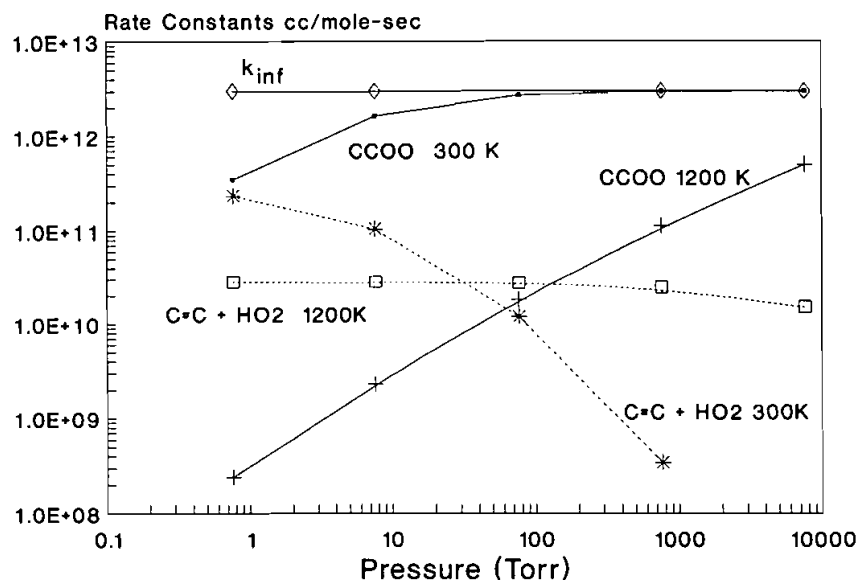


FIGURE 3 Predicted effect of temperature and pressure on the various channels for the $C_2H_5 + O_2$ reaction. k_{inf} [$\diamond$] is the high-pressure addition rate constant. Rate constants for production of the stabilized adduct C_2H_5OO are shown at 300 K [$\blacksquare$] and 1200 K [$+$]. Rate constants for production of $C_2H_4 + HO_2$ are shown at 300 K [$*$] and 1200 K [$\square$].

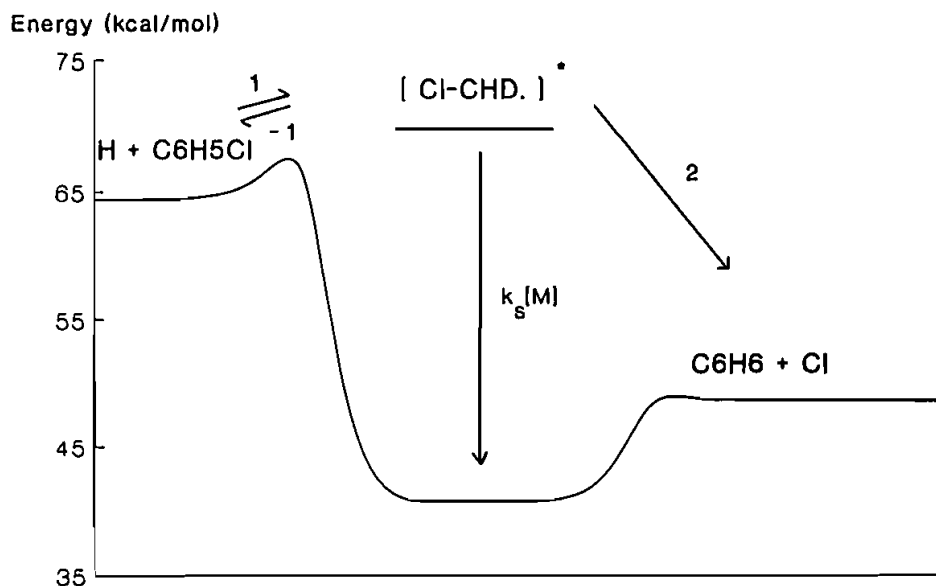
Comparisons of ethyl radical and ethylene concentrations with the low pressure data of Slagle *et al.* (1984; 1986) and the room temperature, but wide pressure range, data of Kaiser *et al.* (1989; 1990) are illustrated in Bozzelli *et al.* (1990). A comparison of the ethylene yield versus pressure over wide temperature ranges (298–900 K) shows excellent agreement. We refer the reader to Bozzelli *et al.* (1990) for additional discussion and interpretation of results from this reaction system.

$H + C_6H_5Cl \leftrightarrow \text{Chloro-cyclohexadienyl Radical}^* \rightarrow \text{Products}$

The second example reaction is the addition of atomic hydrogen to the carbon of the C–Cl bond (ipso addition) in chlorobenzene. An energy diagram for the reaction is shown in Figure 4 and Table IV lists the input parameters for the CHEMACT calculation. The addition of H at the ipso position forms an energized chloro cyclohexadienyl radical ($ClCyCHD^*$) with approximately 32.5 kcal/mole of excess energy relative to the stabilized $ClCyCHD'$. ΔH_f^0 of $ClCyCHD'$ is determined from the parent chloro-cyclohexadiene and the bond energy for the doubly allylic C–H bond of Tsang (1986). The energized complex has three reaction channels available to it, within the thermodynamic energy range shown in Figure 4:

- dissociation to lower energy products, $Cl + \text{benzene } (C_6H_6)$,
- dissociation back to reactants,
- stabilization by collisions with the bath gas.

Results from the CHEMACT calculation illustrate that this hydrogen atom addition to the Cl-carbon in chlorobenzene forms an energized complex which rapidly dissociates to the low energy exit channel, $Cl + C_6H_6$, for all pressures below 10 atm. We chose this reaction because it is an example where the rate constant has virtually no pressure dependence—even though the process is not an elementary reaction. The

FIGURE 4 Potential energy diagram for the reaction of $\text{H} + \text{C}_6\text{H}_5\text{Cl}$.

apparent rate constant for reaction to benzene + Cl is, therefore, simply the rate constant for formation of the adduct ($k = 1.5 \times 10^{13} \exp(-7.5 \text{ kcal/mol}/RT)$). This number, which was obtained from our previous experimental study (Ritter *et al.*, 1990) on chlorobenzene reaction in a hydrogen atmosphere compares favorably with a recent measurement by Cui *et al.* (1989) who report $k = 2.2 \times 10^{13} \exp(-9 \text{ kcal/mole}/RT)$ and of Louw and Mulder (1988), who report a rate constant of $7.4 \times$

TABLE IV
Input parameters for $\text{C}_6\text{H}_5\text{Cl} + \text{H} = [\text{ClCHD}\cdot]^{\#} = \text{C}_6\text{H}_6 + \text{Cl}$
(cf. Figure 4 for Potential Energy Diagram)

k	A (s^{-1} or $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$)	E_a (kcal/mole)	Reference
1	$1.5E + 13$	7.5	a
-1	$3.6E + 12$	31.0	a
2	$1.2E + 13$	8.9	b
	$\langle v \rangle = 1097 \text{ cm}^{-1}$		c
	$\sigma = 5.69 \text{ \AA}$		d
	$\epsilon/k_B = 526.9 \text{ K}$		d

^aRitter, Bozzelli and Dean (1990). Reverse reaction (k_{-1}) from thermodynamics. Alternate values for k_1 could come from data of Cui *et al.* (1989) or data of Louw and Mulder (1988). k_{-1} would then come from microscopic reversibility.

^b A_2 and E_2 based upon reverse reaction with $A_{-2} = 2.0E + 13 \text{ cm}^3 \text{mol}^{-1} \text{s}^{-1}$, $E_{-2} = 0.5 \text{ kcal/mole}$, estimated from the average value of Cl atom addition to olefins and monochloroolefins (Kerr *et al.*, 1981).

^cEstimated by fitting the heat capacity estimates from group additivity to a harmonic oscillator model for vibrational contributions (Ritter, 1989).

^dEstimated as described in the text.

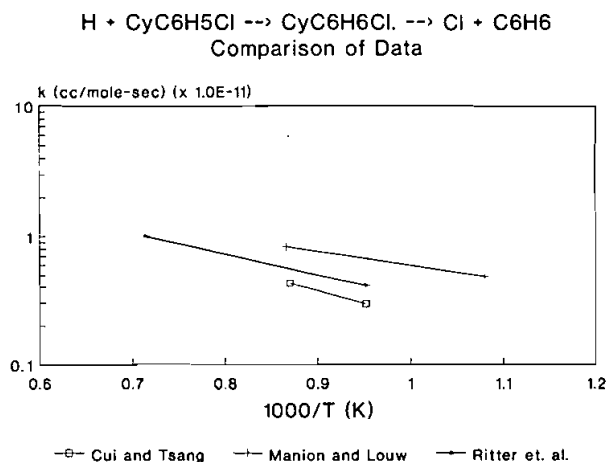
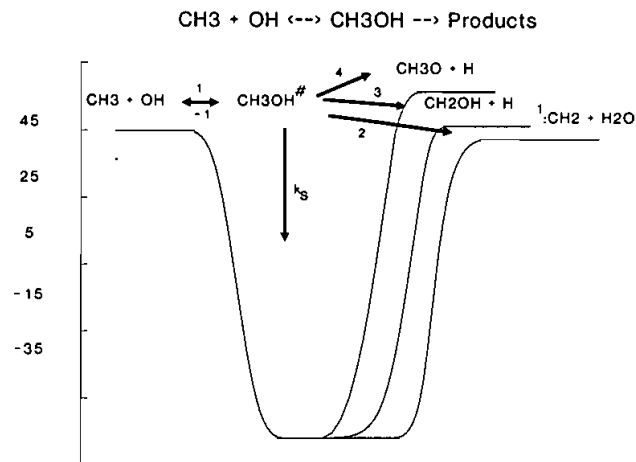


FIGURE 5 Comparison of the QRRK predictions for the production of $\text{Cl} + \text{C}_6\text{H}_6$ from the reaction $\text{H} + \text{C}_6\text{H}_5\text{Cl}$ with recent measurements. [+] from Ref. 65 [□] from Ref. 63.

$10^{12} \exp(-5.02 \text{ kcal/mole}/RT)$. Figure 5 shows a comparison of the calculated rate constants versus $1/T$ using the above input parameters with the data of Cui *et al.* and of Louw and Mulder. There is no pressure dependence in the calculations for temperatures above 1000 K until nearly 10 atm, since all of the reaction goes via the low energy exit channel. Varying the bath gas will also not have any effect in this pressure and temperature range.

$\text{CH}_3 + \text{OH}$

The last reaction we use to illustrate the CHEMACT code is the combination reaction of OH with methyl radical. We chose this because it is a combination reaction, rather than addition to an unsaturated bond, and it has a much deeper well than the previous two examples. It also provides an example of a simple unimolecular dissociation of the adduct as well as a more complex reaction—a concerted elimination. We note that there is sufficient uncertainty in the literature data for Warnatz (1983) to omit recommending a rate constant for this reaction. Here we utilize the similarity of the reaction to $\text{CH}_3 + \text{CH}_3$ to calculate apparent rate constants over a wide pressure and temperature range. A more extensive treatment is available (Dean *et al.*, 1987). A potential energy level diagram of the reaction system is illustrated in Figure 6 and the input parameters for the calculation are listed in Table V. The $\text{CH}_3\text{OH}^\#$ adduct may undergo simple cleavage of the newly formed C–OH bond (k_{-1}), cleavage of either the C–H or O–H bonds (k_3 or k_4), in addition to a concerted H_2O elimination reaction (k_2). The $\text{CH}_3\text{–OH}$ bond is the lowest energy of the simple fission channels, and this channel has the highest Arrhenius A -factor because of entropic effects. The $\text{C}^\# \text{H}_2\text{OH} + \text{H}$ channel is the next lowest energy simple unimolecular reaction, but this has a lower A -factor (relative to C–OH cleavage) due to the lower entropy of H relative to OH. The $\text{CH}_3\text{O} + \text{H}$ (k_4) channel is the highest energy reaction channel. This channel is not considered in the CHEMACT output presented, because the relatively low Arrhenius A -factor combined with the high E_a make it unimportant. The concerted elimination of H_2O is the lowest energy reaction path, but the lower

FIGURE 6 Potential energy diagram in kcal/mole for the reaction of $\text{CH}_3 + \text{OH}$.

A-factor here (estimated from measurements of the reverse reaction of $^1\text{CH}_2$ insertion into H_2O) makes this channel less important than reaction to $\text{H} + \text{C}^1\text{H}_2\text{OH}$. (Another possible channel is the concerted elimination of H_2 , with CH_2O as the other product. Although these products are much lower in energy than the reactants, there is expected to be a very high barrier for this process (Dean *et al.*, 1987).) Figure 7 shows the predicted values for stabilization and the three major reaction channels. Stabilization is the most important channel at 1 atm pressure in N_2 below 1700 K, but the $\text{H} + \text{C}^1\text{H}_2\text{OH}$ and $^1\text{CH}_2 + \text{H}_2\text{O}$ channels have large rate constants, and thus could be quite important in many combustion systems. Figure 7 also illustrates the importance of fall-off effects at higher temperatures. Although Tsang and Hampson (1986) suggest an overall rate constant, *i.e.*, to all product channels, of $5.4 \times 10^{13} \text{ cm}^3 \text{ mole}^{-1} \text{ s}^{-1}$ between 300–2500 K, the QRRK calculations clearly show the onset of falloff at temperatures as low as 700 K at 1 atm in N_2 . At 2100 K, the total rate

TABLE V
Input parameters for $\text{CH}_3 + \text{OH}$
(*cf.* Figure 6 for Potential Energy Diagram)

k	A (s^{-1} or $\text{cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$)	E_a (kcal/mole)	Reference
1	$2.6E + 13$	0.0	a
-1	$6.3E + 15$	91.8	a
2	$4.4E + 14$	89.8	b
3	$1.3E + 15$	93.3	c
4	$2.4E + 14$	103.4	c
	$\langle v \rangle = 1519 \text{ cm}^{-1}$		d
	$\sigma = 3.78 \text{ \AA}$		e
	$\epsilon/k_B = 377 \text{ K}$		e

^a k_1 from $\text{CH}_3 + \text{CH}_3$ combination (Allara and Shaw, 1980) k_{-1} from microscopic reversibility.

^b A_2 and E_2 based upon reverse reaction, using data of Hatakeyama *et al.* (1981).

^c k_3 and k_4 based upon $k_{\text{rev}} = 6.4E + 13$.

^dBurcat (1984).

^eBen-Amotz and Herschbach (1990).

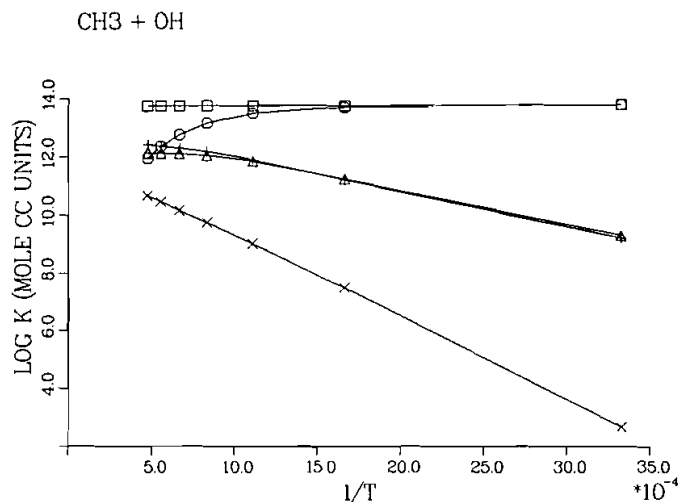


FIGURE 7 QRRK predictions for the various channels of CH₃ + OH as a function of temperature at 1 atm in N₂. [□] is the high pressure recombination rate constant. [○] is the rate constant for production of the stabilized adduct CH₃OH. [Δ] is the rate constant for production of ¹CH₂ + H₂O. [+] is the rate constant for production of C + H₂OH + H. [×] is the rate constant for production of CH₃O + H.

constant is down by nearly a factor of 5. At these temperatures the reverse reaction, by which the energized adduct dissociates back to CH₃ + OH, begins to dominate, substantially reducing the overall rate to the product channels.

USE IN DETAILED MODELING CALCULATIONS

CHEMACT provides a method to estimate apparent bimolecular rate constants for the various products that can result from a chemically activated reaction. Since these rate constants depend on both pressure and temperature, it is important that the user include the appropriate expression for the specific pressure of interest in the modeling calculation. (This expression is then valid over whatever temperature range was fitted with CHEMACT.) A method which allows the pressure dependence to be described in such a way that it is compatible with the new version of CHEMKIN (Kee *et al.*, 1989) is under development (Dean *et al.*). In addition, the user must account for further reactions of the stabilized adducts, which are products of the reaction. The dissociation of the stabilized adduct back to reactants is automatically included if the reactions are written reversibly, but, for completeness, dissociation to the other possible products must be explicitly included. Of course, these rate constants are also generally pressure-dependent so some type of fall-off estimate is required. We account for this via a program named DISSOC, which is considered separately (Bozzelli *et al.*), but any other approach which accounts for dissociation of thermalized species could be used. Examples include use of the Kassel integral (Emanuel, 1969) as well as the methods described in Weston (1986).

CONCLUSION

The examples provided illustrate that the calculation of rate parameters for addition, combination and insertion reactions with reasonable account for fall-off can be

readily performed with the CHEMACT code. The input parameters are obtainable to a reasonable degree of accuracy to allow use of the code for estimating the apparent rate constants as functions for both temperature and pressure for improved accuracy in construction of detailed kinetic models. The code, in executable format for an IBM PC and compatible systems, is obtainable by writing the authors.

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APPENDIX A

DETERMINATION OF KINETIC PARAMETERS BY MICROSCOPIC REVERSIBILITY

Calculation of a reverse rate constant, k_r , for an elementary reaction from the forward rate constant, k_f , utilizes the relationship:

$$k_f/k_r = K_c,$$

where K_c is the equilibrium constant in concentration units. The equilibrium constant K_p (pressure units) for a given reaction can be calculated if the free energy of the reaction is known:

$$K_p = \exp(-\Delta G_{rxn}/RT)$$

Note that $K_c = K_p$ only when the number of moles of product equals the number of moles of reactant, i.e., when Δn is 0. When Δn is not equal to 0, K_c can be obtained from use of the ideal gas law:

$$\ln(K_c) = -\Delta G_{rxn}/RT - \Delta n \ln(R'T),$$

where $R = 1.987 \text{ cal}/(^{\circ}\text{K-mole})$ and $R' = 82.06 \text{ cm}^3\text{-atm}/(^{\circ}\text{K-mole})$ when the concentration is expressed in moles/cm³. Using standard thermodynamic relationships, this can be written as:

$$\ln(K_c) = -(\Delta E_{rxn} + \Delta nRT - T\Delta S_{rxn})/RT - \Delta n \ln(R'T).$$

Rearranging:

$$\ln(K_c) = -\Delta E_{rxn}/RT + \Delta S_{rxn}/R - \Delta n(\ln(R'T) + 1).$$

But

$$\ln K_c = \ln(k_f/k_r) = \ln(A_f/A_r) - (E_f - E_r)/RT,$$

where A_f is the Arrhenius pre-exponential factor and E_f is the activation energy for the forward reaction, the A_r and E_r are the corresponding terms for the reverse reaction. So we can identify

$$\ln(A_f/A_r) = \Delta S_{rxn}/R - \Delta n[\ln(R'T) + 1],$$

$$E_f - E_r = \Delta E_{rxn} = \Delta H_{rxn} - \Delta nRT.$$

Golden (1971) discusses these calculations in more detail.

APPENDIX B

CHEMACT THERMO DATA

Species	ΔH_f (298)	S (298)	C_p 300	400	500	600	800	1000	1500
HO ₂	3.50	54.73	8.44	8.98	9.48	9.44	10.74	11.37	12.40
³ CH ₂	92.37	46.33	8.30	8.73	9.16	9.58	10.38	11.09	12.41
¹ CH ₂	101.45	44.15	8.30	8.73	9.16	9.58	10.38	11.09	12.41
CH ₃	34.82	46.38	9.28	10.07	10.85	11.59	12.96	14.16	16.30
CH ₃ O [•]	3.96	54.65	9.04	10.66	12.21	13.66	16.24	18.34	21.61
C [•] H ₂ OH	-6.10	56.30	11.59	13.13	14.50	15.73	17.76	19.32	21.77
CH ₂ OH	-48.06	57.30	10.75	12.83	14.75	16.52	19.59	22.05	25.98
C ₂ H ₄	12.54	52.39	10.41	12.84	14.99	16.88	20.00	22.39	26.17
C ₂ H ₅	28.30	59.29	12.11	14.91	17.39	19.58	23.20	25.98	30.41
CCOO [•]	-4.70	73.75	18.07	22.10	25.59	28.60	33.40	36.89	42.03
CCO [•]	-4.22	64.52	14.00	17.12	20.01	22.66	27.24	30.87	36.45
C [•] COOH	7.24	79.77	19.56	23.81	27.40	30.42	35.06	38.28	42.81
CLBZ	12.70	75.19	23.73	30.69	36.40	41.06	47.95	52.57	59.06
cyC ₆ H ₆	19.80	64.26	19.61	26.71	32.67	37.66	45.29	50.61	58.12
cyC ₆ H ₇	49.86	72.03	20.88	28.65	35.15	40.56	48.74	54.35	62.25